Human-Environmental Interaction

RRHS APHG
2019-2020

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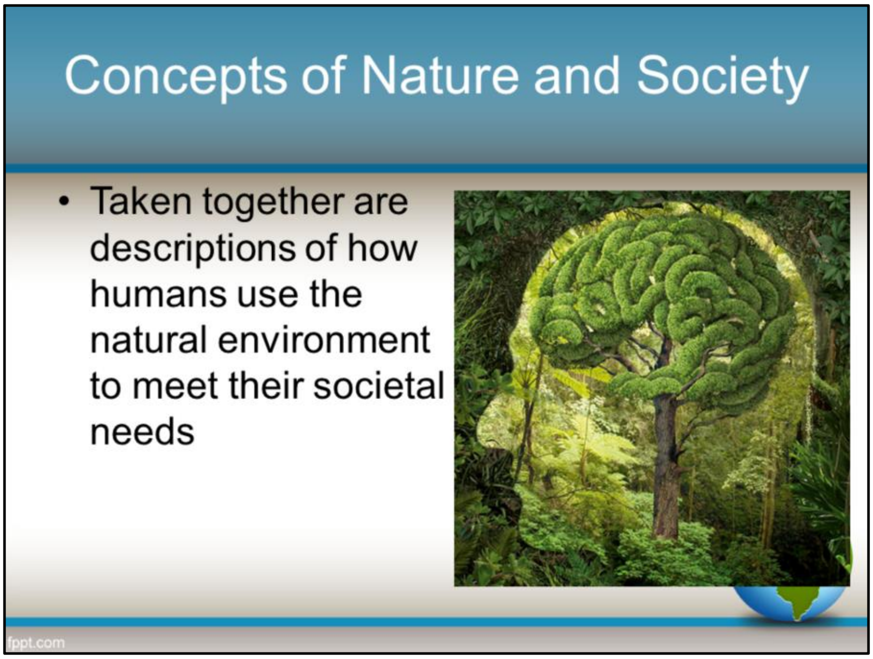


Society is the collective group of people living together in a more or less ordered community. We live and learn the norms (rules, both written and unwritten) of a society. Societies began as the earliest tribes of nomadic herders moved across the continents to populate the world. As people settled and created civilizations (and later towns and cities), society became much much more complex.

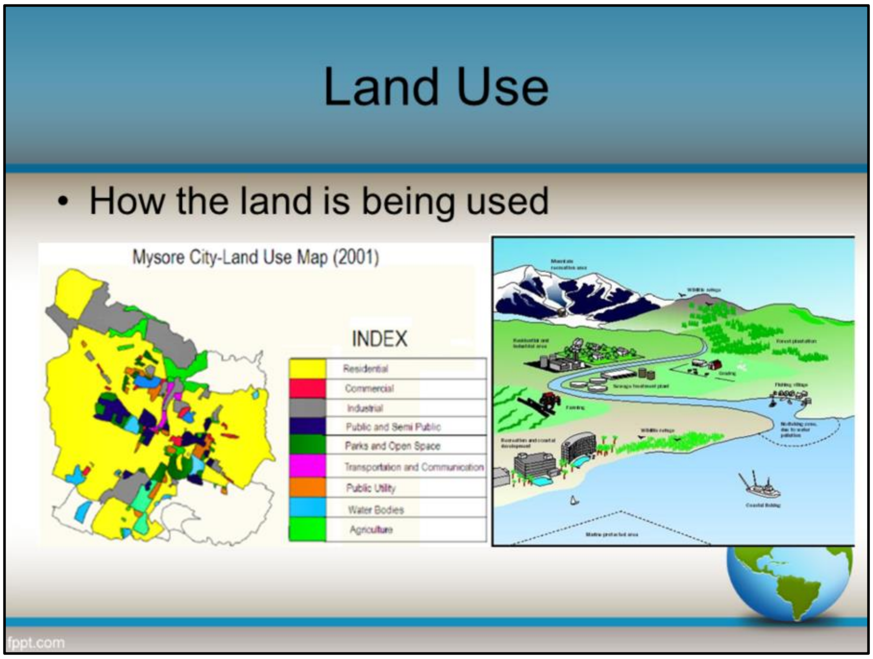
Nature in the broadest sense, is the natural, physical, or material world or universe. "Nature" can refer to the phenomena of the physical world, and also to life in general.

Society NEEDs nature to survive, and in turn, society also impacts nature. We use the natural resources, we modify the environment to fit our needs, and we adapt our practices and activities to function within nature at a given place.

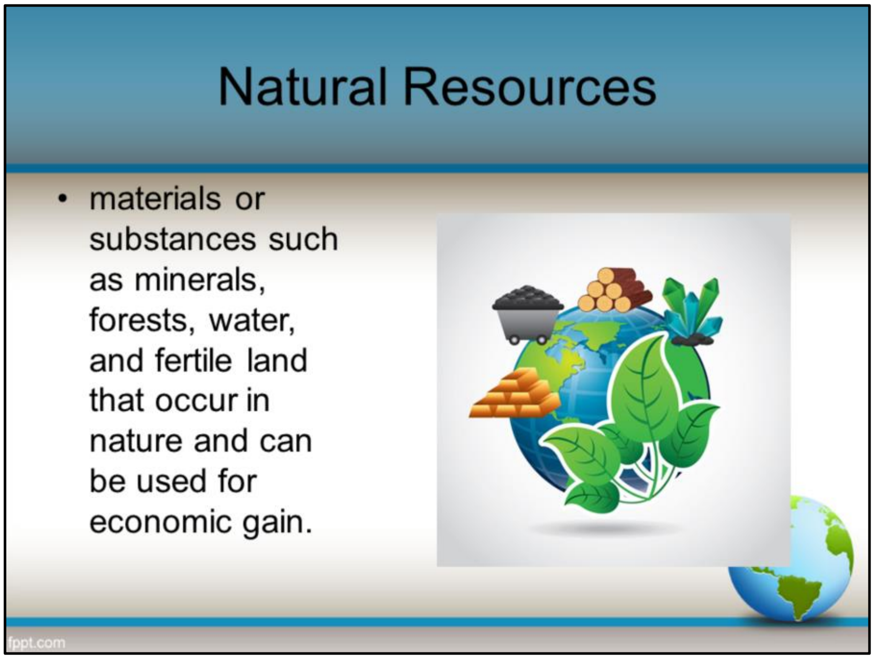
Whether it’s a city with skyscrapers, terraces carved into a hillside for farming, or a footpath through a grassy field, if humans have touched it, it is part of the cultural landscape.

Cultural landscape = built landscape (even if it’s not a building, humans have changed it in some way through use).

The concept of cultural landscape was first presented by Carl Sauer (yes, you should know this name in association with cultural landscape)



Different types of land uses include residential (housing), commercial (businesses), industrial (factories), public (parks and/or government owned), Transportation (roads, railroads, etc.), public utility (electric, water treatment, etc), water bodies (natural or man-made), or agricultural (farms and ranches).



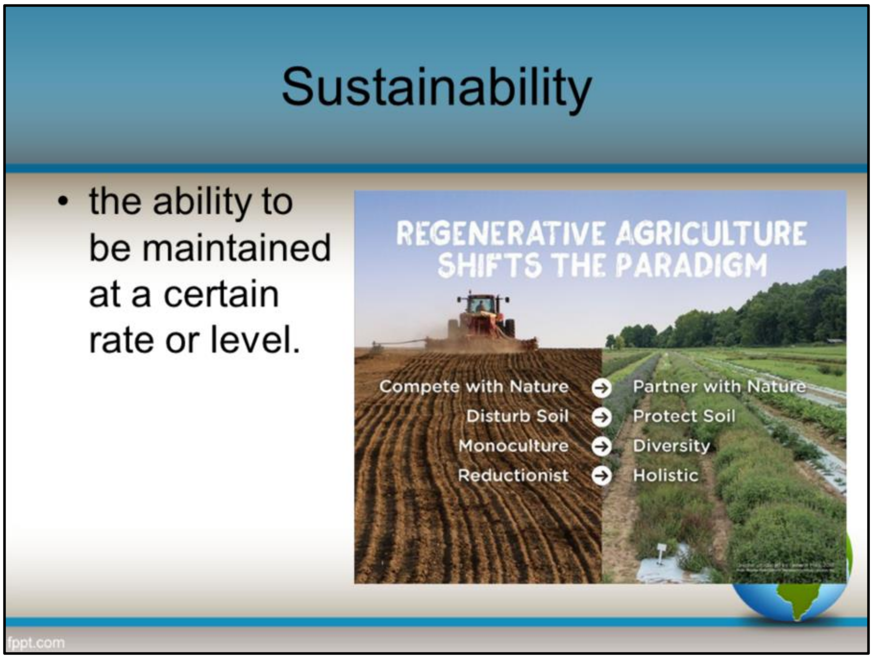
Two of the most important natural resources we will discuss in this class are fertile land (arable land) and water. Without these two natural resources, civilizations cannot survive in that place.

Other import natural resources include fossil fuels, which are used to power industry, our homes, and our vehicles. These include coal, oil, and natural gas. The most economically advanced countries consume the most fossil fuels.

Renewable resources: these include sunlight, hydropower (power produced by damming a waterway), and wind.

Minerals: iron ore (used to make steel), gold, silver, copper (used to make electrical wiring), mercury, diamonds, bauxite (used to make aluminum), etc.

Typically, countries that do a lot of manufacturing will need large amounts of minerals and power from fossil fuels or other sources of energy to make their products. If the country does not have them on their own land, they must be able to acquire them from another place. This need for resources in manufacturing has been very significant in terms of global trade, politics, and the spread of cultures.



Sustainability is most often defined as meeting the needs of the present without compromising the ability of future generations to meet theirs.

This means that sustainability goals are meant to allow the use of a resource, but at a rate that will protect it for future generations.

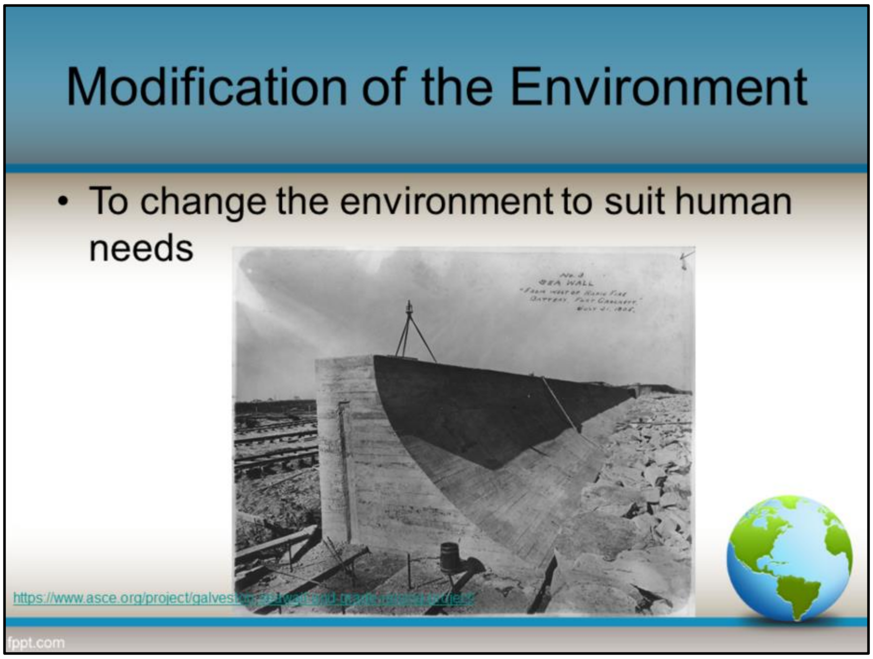
Think about the fishing industry. Fish must reproduce to add new fish to the population. If the fishing companies over-fish, that means they are catching too many fish for the remaining population to replenish, and in the future when they go to fish, they may be out of luck. But if these companies follow sustainability guidelines, they can catch a certain number of fish while leaving enough behind to reproduce and repopulate the area so that they can go fishing again in the future.

Sustainable agriculture seeks to limit the destruction that can be caused by intensive farming, such as soil depletion and erosion, chemical pollution of the water, invasive species being introduced or beneficial species being harmed. In our agriculture unit, we will look at the effects of different methods of farming on the environment.

Sustainability can be very personal as well. If you are overcommitted or overworked to the point that you are sacrificing sleep, not eating well, and just generally running yourself ragged, you will soon burn out because such a pace is unsustainable. Pace yourself. Give yourself time to recoup, get plenty of sleep, and make sure you're taking care of yourself so that you are able to do well in the future.

Natural disasters are natural events such as a flood, earthquake, or hurricane that causes great damage or loss of life. Environmental phenomenon are only considered hazardous (or disasters) if people are impacted. So if a hurricane hits an unpopulated island, it is not considered a disaster, but if it hits a populated place it is – and some of the hardest hit areas are also some of the most densely populated (such as the Philippines, the Texas Gulf Coast, etc.).

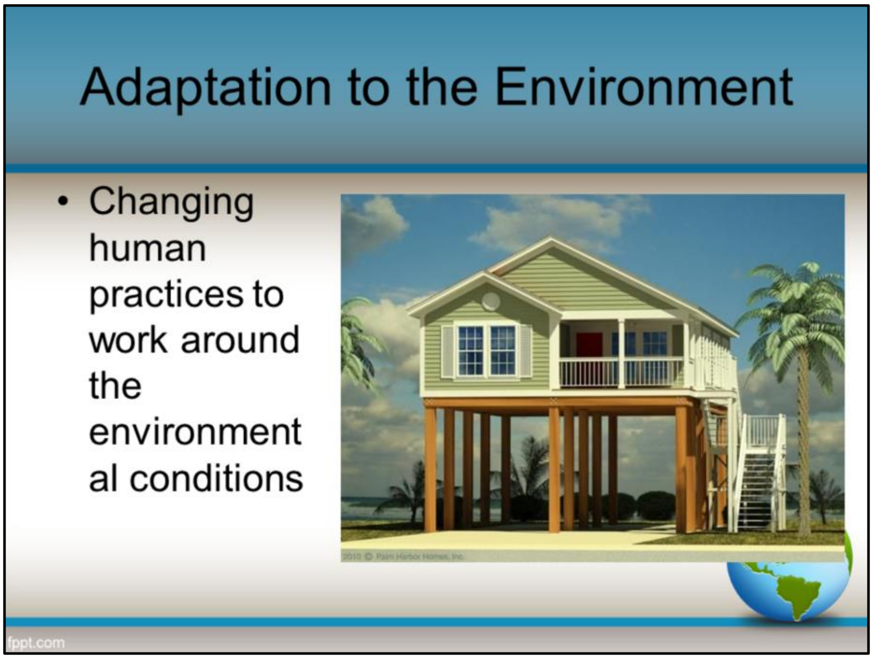
Technological disasters occur when human-built modifications to the environment fail, such as a dam failure (image on right), an oil rig explosion/spill in ocean (images on left), or a levee break during a flood. Usually there is significant damage to property and/or loss of both human and wildlife when such events occur.



Modifications of the environment occur when people change the physical environment to suit their own desires, needs, and economic activity.

Modifications include clearing land for farming, damming a river for flood control or to create a water reservoir for irrigation and drinking water needs, channelizing rivers (re-directing them to a fixed location instead of letting them flow and move on their own), building islands/raising the grade on islands (Galveston example below), digging canals and channels for transportation, or any other human activity that changes the natural environment.

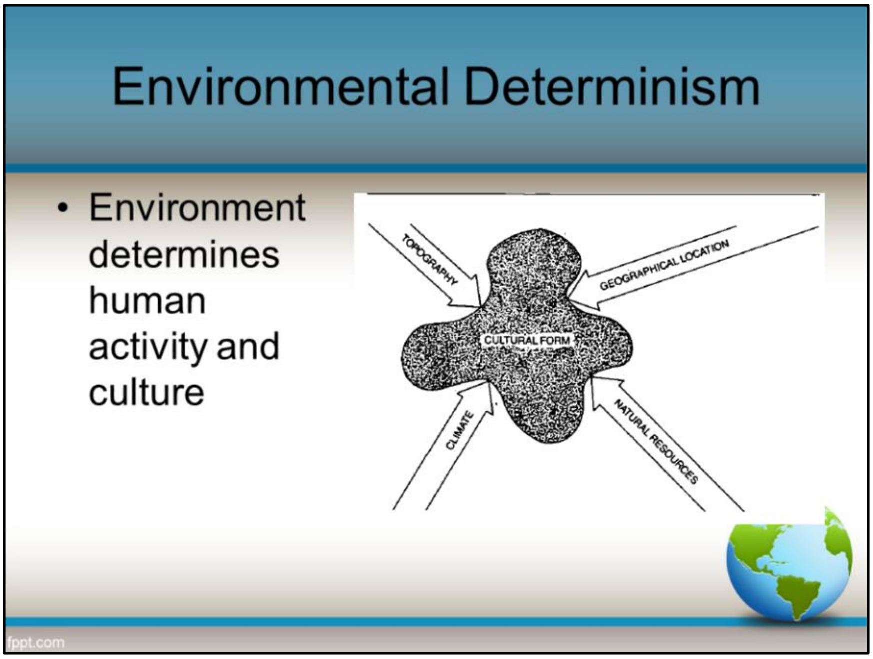
The image above was taken during the construction of the seawall on Galveston Island in Texas. The 17 foot wall was built to protect the island from the dangerous storm surges (raised ocean level) that accompany hurricanes. In 1900, Galveston was hit head-on by the most deadly hurricane to ever hit the United States. It killed an estimated 6000 people and destroyed almost every building on the low-lying barrier island. In response to the disaster, the government built the sea wall and also raised the grade (elevation) of the island….this was a MASSIVE engineering undertaking as every single man-made structure, sidewalk, streetcar line, etc. had to be jacked up, then sand that had been dredged (dug) from Galveston harbor was pumped in. New foundations were built, and the structures set back down. About 500 city blocks were raised using 16.3 million cubic yards of sand spread from a few inches to eleven feet thick. The entire island was modified by raising it to protect it from future natural hazards.



Adaptation, on the other hand, is changing our human practices to live and function in the environment as it exists. For example, instead of building a house on a low foundation in a flood-prone area, the house could be built on stilts so flood waters can ebb and flow under it during flood events. Another example of adaptation is in our clothing – if it's cold, we wear coats. If it's hot, we dress in much cooler clothing and wear hats and sunscreen to protect our skin from the sun.

Theories of Human-Environmental Interaction

- Does the environment shape the culture or does the culture shape the environment?
  - Two schools of thought:
    - Environmental Determinism
    - Possibilism



In this theory, the natural environment is the main determinant of human activity in various places. It influences everything from economic activity (farming, ranching, fishing, etc.) to language, religion, and culture.

This was the predominant school of thought on this matter until the about the 1950s/60s.

Jared Diamond is a leading modern-day influencer in regard to this school of thought. His publications and videos reflect this theory.

An example of an environmental determinism thought is that temperate climates (the mid climates between the hot equatorial zone and the cold polar zones) produce inventive, industrious, and democratic societies that are most likely to take control of other societies in less-superior climate zones.

As a whole, societal understanding of race, ethnicity, and place have changed significantly in the past 100 years or so, and as such this theory is now largely considered to be defunct.

Possibilism is the modern-day leading scholarly school of thought in terms of cultural development and exchange, and came about as a result of the industrial revolution.

Think about it – all of the new technologies and innovations, practices, and scientific discovery that happened as a result of the industrial revolution led to a significant shift in way of life. It enabled people to live in any environments and modify/make adaptations to their activities in order to function in any place. We can irrigate deserts into farms. We can use air-conditioning to live comfortably in Houston (or Round Rock). We have the technology we need to alter the environment to suit our needs.

Possibilism = with technology, anything is possible in any location.

The image above shows center-point pivot irrigation in the Sahara Desert.